Year 10 Mathematics

Shape and Space 1

5.1 Triangles

Triangles

STARTER · 5 MIN

Date: _____ **Title:** Triangles

LEARNING OBJECTIVES

- Use the angle sum of a triangle
- Use the properties of isosceles and equilateral triangles
- Use the exterior angle of a triangle

STARTER *Answer in your book. No calculators.*

1. Work out 180 − 65
2. Work out 180 − 72 − 48
3. Work out 90 − 37
4. Solve 2x + 40 = 180

Starter — Answers

STARTER

Tick or correct in green pen

1. $180 - 65 =$ **115**

2. $180 - 72 - 48 =$ **60**

3. $90 - 37 =$ **53**

4. $2x = 140, \quad x =$ **70**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can use the angle sum of a triangle	☐
SILVER	I can use the properties of isosceles and equilateral triangles	☐
GOLD	I can use the exterior angle of a triangle	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Angles in Triangles

INTRODUCTION · 3 MIN

The three angles in any triangle add up to 180°.

Angle sum

$a + b + c = 180°$

$50° + 70° + 60° = 180°$

Isosceles

Two equal sides and two equal base angles.

Base angles are equal

Equilateral

Three equal sides and three equal angles.

Each angle is 60°

Exterior angle

Equals the two opposite interior angles added.

$48° + 67° = 115°$

Examples 1 and 2 — Angle Sum of a Triangle

EXAMPLES · 5 MIN

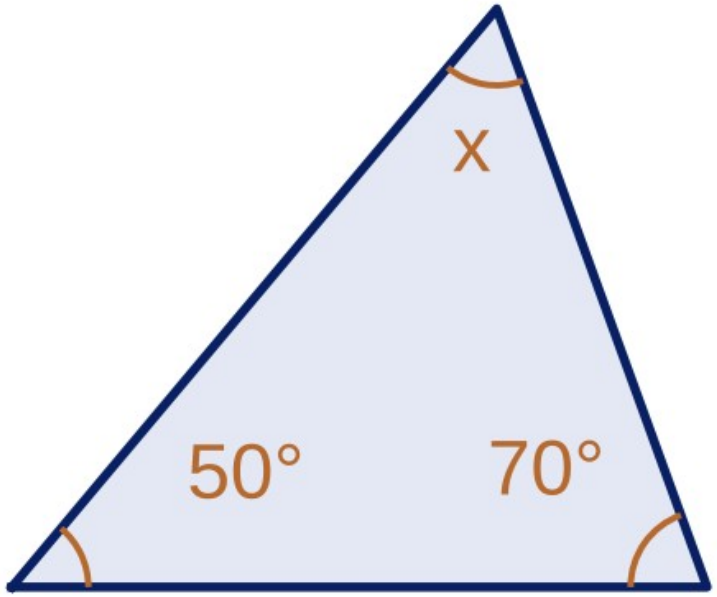
BRONZE

Watch and copy into your book

Hint a + b + c = 180°

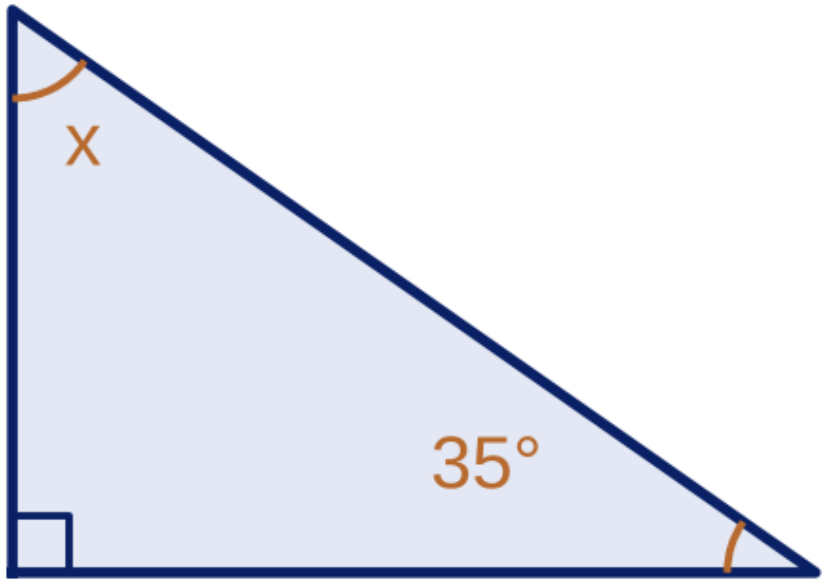
Example 1

Find the size of angle x.



Example 2

Find the size of angle x.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$x = 180 - 50 - 70$

$x = 60°$

Tip: angles in a triangle add up to 180°.

Example 2

$x = 180 - 90 - 35$

$x = 55°$

Tip: the square shows a right angle (90°).

Task 1

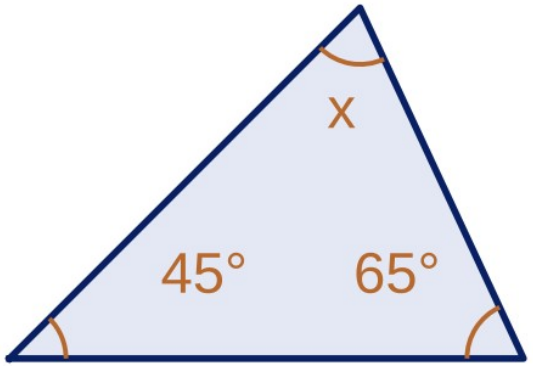
PRACTICE · 9 MIN

BRONZE

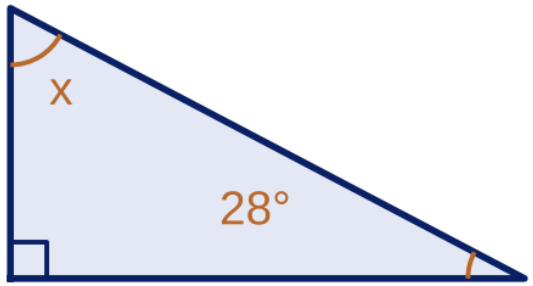
Find the size of each angle marked x

Hint a + b + c = 180°

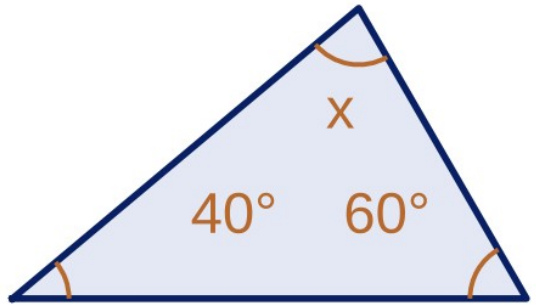
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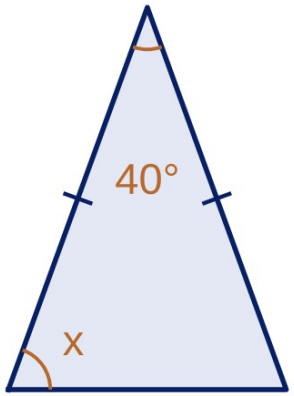
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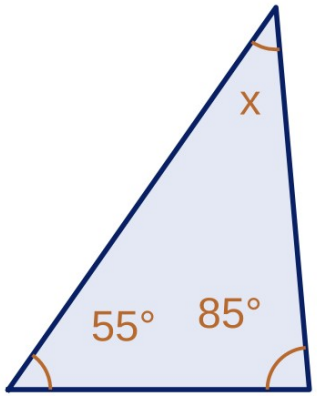
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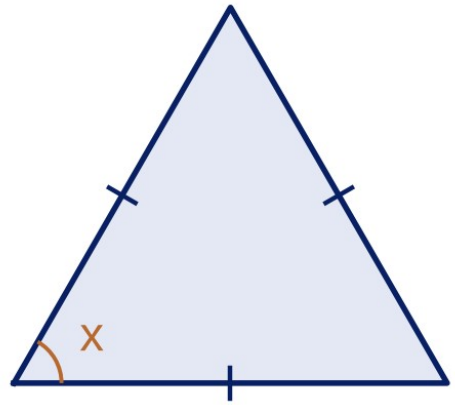
4



5



★



Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $x = 180 - 45 - 65 =$ **70°**

2. $x = 180 - 90 - 28 =$ **62°**

3. $x = 180 - 40 - 60 =$ **80°**

4. $x = (180 - 40) \div 2 =$ **70°**

5. $x = 180 - 55 - 85 =$ **40°**

★ $x = 180 \div 3 =$ **60°**

Examples 3 and 4 — Isosceles and Exterior Angles

EXAMPLES · 5 MIN

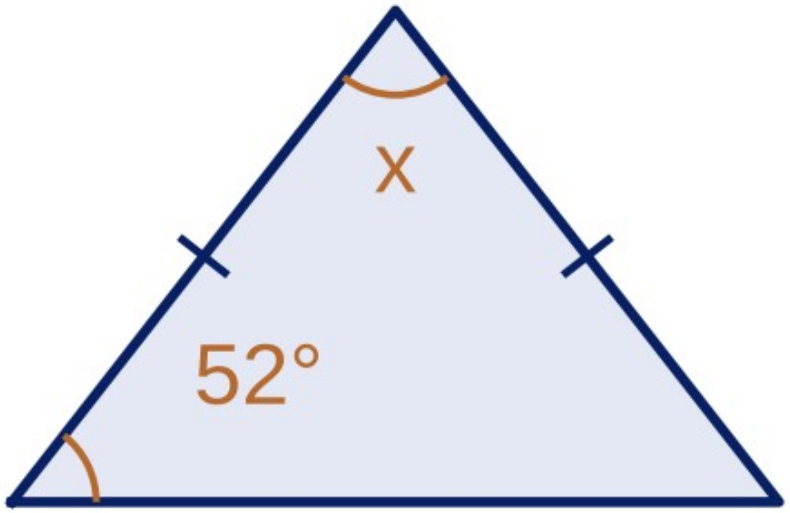
SILVER

Watch and copy into your book

Hint
a + b + c = 180°
exterior angle = sum of interior opposite angles

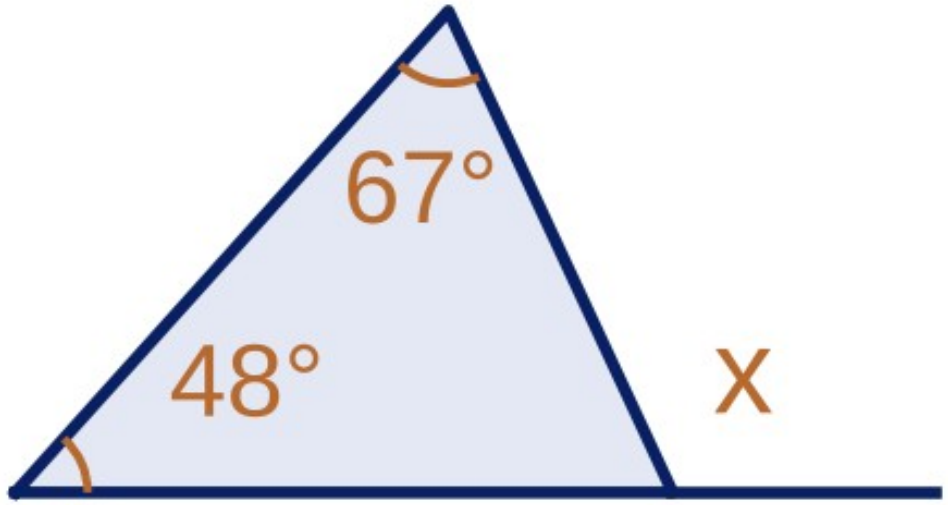
Example 3

The triangle is isosceles. Find angle x.



Example 4

Find the exterior angle x.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$x = 180 - 52 - 52$

$x = 76°$

Tip: the two base angles are equal.

Example 4

$x = 48 + 67$

$x = 115°$

Tip: exterior angle = sum of the two opposite interior angles.

Task 2

PRACTICE · 9 MIN

SILVER *Find the size of angle x*

Hint a + b + c = 180° exterior angle = sum of interior opposite angles

1. An isosceles triangle has base angles of 35°. Find the third angle.

2. An isosceles triangle has a top angle of 100°. Find each base angle.

3. Two interior angles are 55° and 72°. Find the exterior angle opposite them.

4. An exterior angle is 140°. One opposite interior angle is 85°. Find the other.

5. The angles of a triangle are x, 2x and 3x. Find x.

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $180 - 35 - 35 =$ **110°**

2. $(180 - 100) \div 2 =$ **40°**

3. $x = 55 + 72 =$ **127°**

4. $140 - 85 =$ **55°**

5. $6x = 180, \quad x =$ **30°**

Examples 5 and 6 — Angle Problems with Algebra

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint
a + b + c = 180°
exterior angle = sum of interior opposite angles

Example 5

(3 marks)

The angles of a triangle are

x + 20, 2x and 3x – 20.

Find the size of each angle.

Example 6

(3 marks)

An isosceles triangle has a top angle of 2x

and base angles of x + 10.

Find the top angle.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$$x + 20 + 2x + 3x - 20 = 180$$
$$6x = 180$$
$$x = 30$$

Angles: 50°, 60°, 70°

Tip: add the angles and make them equal 180.

Example 6

$$2x + 2(x + 10) = 180$$
$$4x + 20 = 180$$
$$x = 40$$

Top angle **= 80°**

Tip: there are two base angles.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint
a + b + c = 180°
exterior angle = sum of interior opposite angles

1. The angles of a triangle are 3x, 4x and 5x. Find the largest angle. (3 marks)

2. The angles of a triangle are x + 10, x + 30 and 2x. Find x. (3 marks)

3. An isosceles triangle has base angles of 3x and a top angle of x + 40. Find the top angle. (3 marks)

4. An exterior angle is 4x. The opposite interior angles are x + 15 and 2x + 5. Find x. (3 marks)

5. The angles of a triangle are 2x, 3x + 10 and 5x − 30. Show that the triangle is isosceles. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $12x = 180, \quad x = 15, \quad 5x =$ **75°**

2. $4x + 40 = 180, \quad 4x = 140, \quad x =$ **35°**

3. $7x + 40 = 180, \quad x = 20, \quad x + 40 =$ **60°**

4. $4x = 3x + 20, \quad x =$ **20**

5. $10x - 20 = 180, \quad x = 20,$ **Angles: 40°, 70°, 70°**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Two angles of a triangle are 72° and 48°. Find the third angle.

SILVER An isosceles triangle has base angles of 65°. Find the top angle.

GOLD The angles of a triangle are x, x + 30 and 2x. Find x.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $180 - 72 - 48 =$ **60°**

SILVER $180 - 65 - 65 =$ **50°**

GOLD $4x + 30 = 180, \quad 4x = 150, \quad x =$ **37.5°**

Exam Practice — Angles in a Triangle

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

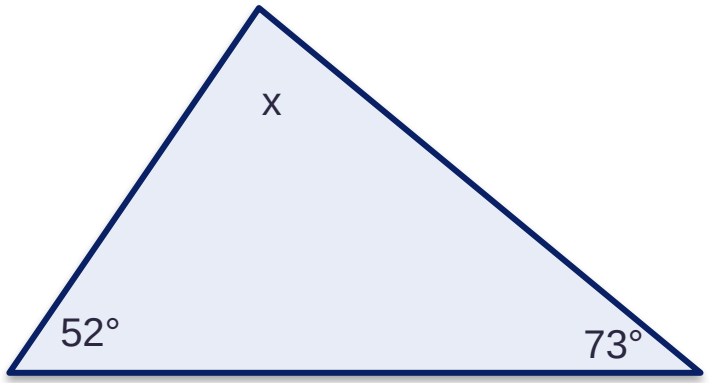
Hint a + b + c = 180°

Question 1

(2 marks)

Work out the size of angle x.

Diagram NOT accurately drawn



..... °

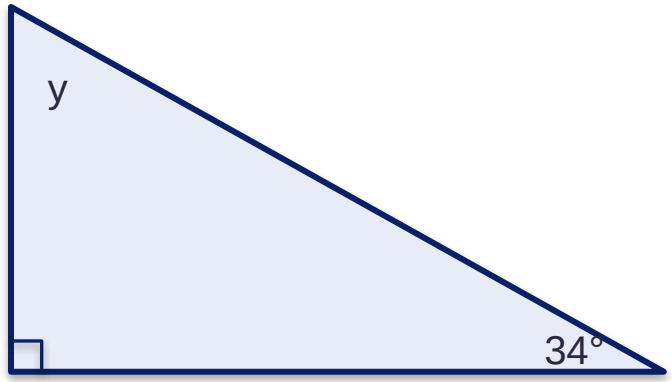
Question 2

(2 marks)

Work out the size of angle y.

Give a reason for your answer.

Diagram NOT accurately drawn



..... °

Exam Practice — Isosceles and Exterior Angles

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

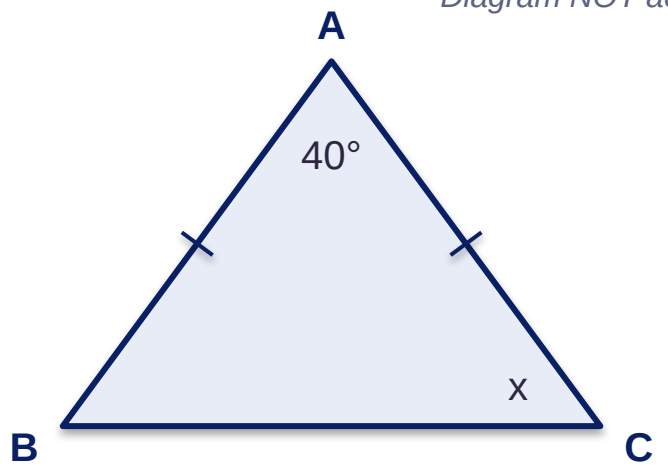
Hint
a + b + c = 180°
exterior angle = sum of interior opposite angles

Question 3

(2 marks)

Triangle ABC is isosceles with AB = AC.
Work out the size of angle x.

Diagram NOT accurately drawn



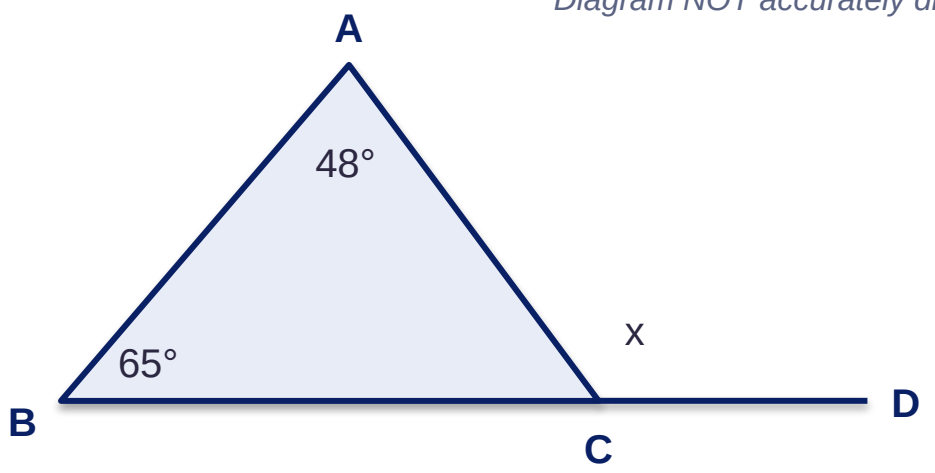
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Question 4

(3 marks)

BCD is a straight line.
Work out the size of angle x.
Give a reason for your answer.

Diagram NOT accurately drawn



..... °

Exam Practice — Algebra with Angles

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint
a + b + c = 180°
exterior angle = sum of interior opposite angles

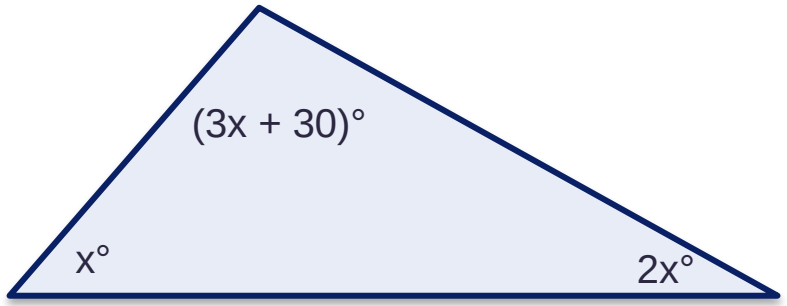
Question 5

(4 marks)

The angles of a triangle are x°, 2x° and (3x + 30)°.

Work out the size of the largest angle.

Diagram NOT accurately drawn



..... °

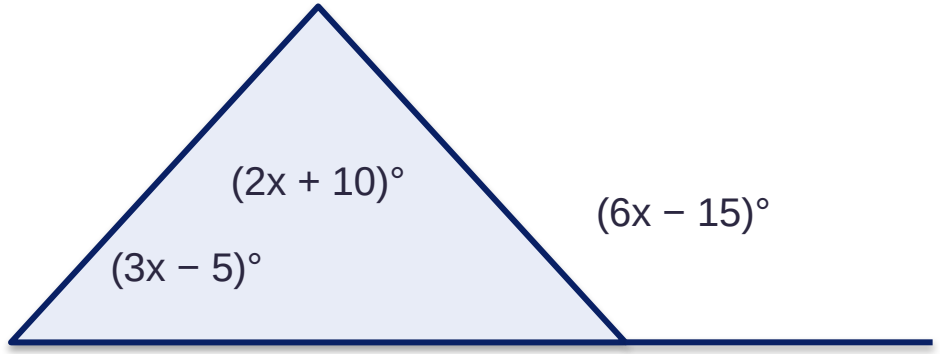
Question 6

(3 marks)

The diagram shows a triangle with one side extended.

Work out the value of x.

Diagram NOT accurately drawn



x = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

180 − 52 − 73 — M1

x = 55° — A1

Tip: angles in a triangle add to 180°.

Question 2

180 − 90 − 34 — M1

y = 56° — A1

(angles in a triangle sum to 180°)

Tip: always write the reason in full.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(180 − 40) ÷ 2 — M1

x = 70° — A1

Tip: base angles of an isosceles triangle are equal.

Question 4

48 + 65 — M1

x = 113° — A1

Exterior angle = sum of the two interior opposite angles — B1

Tip: or 180 − (180 − 48 − 65) = 113.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$x + 2x + 3x + 30 = 180$ — M1

$6x = 150,\ \ x = 25$ — M1

$3(25) + 30 = 105°$ — A1

Check: $25 + 50 + 105 = 180$ ✓

Tip: find x, then answer the question asked.

Question 6

$6x - 15 = 2x + 10 + 3x - 5$ — M1

$6x - 15 = 5x + 5$ — M1

$x = 20$ — A1

Tip: exterior = sum of interior opposite angles.